SupplyChain360

Data Platform

End-to-End Data Engineering Pipeline

- Airflow
- Snowflake
- dbt
- Terraform
- AWS
- Docker
- GitHub Actions

1.4M+
rows ingested

49
dbt tests passing

7
Airflow tasks

Mohammed | April 2026

THE BUSINESS PROBLEM

SupplyChain360 manages product distribution across hundreds of US retail stores — but was flying blind on operations.

Stockouts

Stores ran out of stock with no early warning system

Delayed Shipments

No visibility into late deliveries or carrier performance

Siloed Data

Data scattered across S3, Google Sheets & Supabase

No Analytics

Manual reporting — no real-time business insights

PLATFORM ARCHITECTURE

DATA SOURCES	INGESTION	STORAGE	TRANSFORM	INFRA / CI-CD
AWS S3 (CSV/JSON)	Apache Airflow	Parquet → S3	dbt Staging	Terraform IaC
Google Sheets (Stores)	Docker Container	Snowflake RAW	dbt Dimensions	GitHub Actions
Supabase PG (7 daily tables)	Daily Schedule	SSM Secrets	dbt Facts	Docker Hub

THE DATA PIPELINE — 7 TASKS

1

`check_idempotency`

Verifies 31 parquet files exist before running

2

`migrate_s3_files`

17 CSV/JSON → Parquet with timestamps

3

`migrate_google_sheets`

800 store records from Sheets API

4

`migrate_supabase`

7 daily sales tables + 6 other tables

5

`load_to_snowflake`

All Parquet → Snowflake RAW schema

6

`data_quality_checks`

5 DQ checks: nulls, dupes, negatives

7

`run_dbt_models`

14 models + 49 tests via dbt

Orchestrated by Apache Airflow · Runs @daily · Retry on failure · Failure alerts enabled

DBT DATA MODELING

STAGING LAYER

7 views — clean & standardize

- stg_products
- stg_suppliers
- stg_warehouses
- stg_stores
- stg_inventory
- stg_shipments
- stg_sales (union 7 daily)

DIMENSIONS

4 tables — reference data

- dim_products
- dim_suppliers
- dim_warehouses
- dim_stores

FACTS

3 incremental tables

- fct_sales (1.4M rows)
- fct_shipments (50K rows)
- fct_inventory (8K rows)

✅ **49/49 dbt Data Quality Tests Passing | Incremental models for fct_sales & fct_shipments**

KEY BUSINESS INSIGHTS

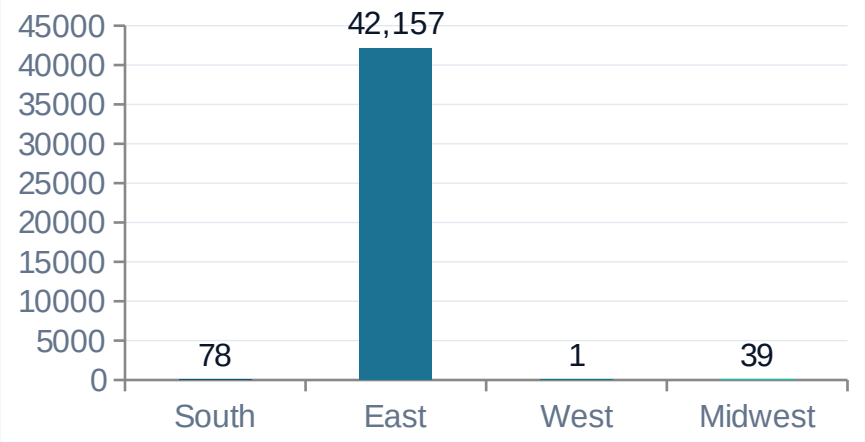
$198M
Total Revenue
March 2026

1.4M
Sales
Transactions

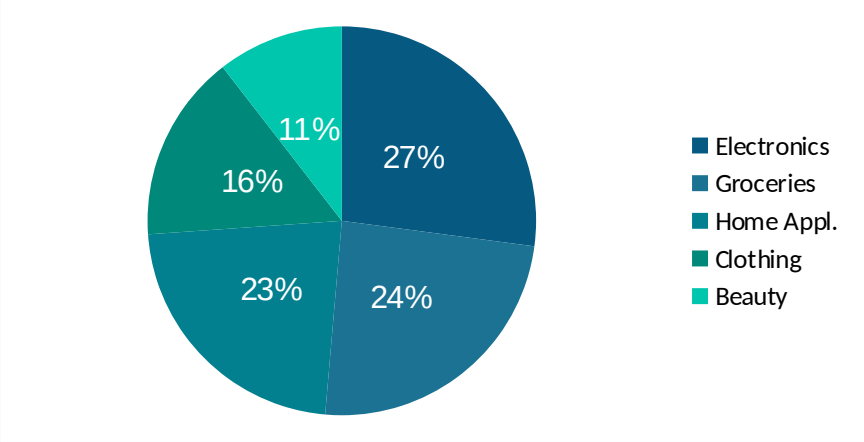
~50%
Late Shipment
Rate (all carriers)

20+
Products Below
Reorder Threshold

Revenue by Region



Revenue by Category



INFRASTRUCTURE & CI/CD

Terraform IaC

- S3 buckets (source + parquet + state)
- Snowflake DB, schemas & warehouse
- AWS SSM Parameter Store (all secrets)
- S3 remote backend for state

GitHub Actions CI/CD

- flake8 Python linting on every push
- sqlfluff SQL linting for dbt models
- Docker build & push to Docker Hub
- dbt test — 49 quality tests on push

Docker + Quality

- Custom image: mdethereal/supplychain360
- All dependencies baked in
- 5 Airflow DQ checks per run
- Idempotency check as first task

TECHNOLOGY STACK

Orchestration

Apache Airflow 2.8.1

7-task DAG · daily · idempotent · retry + alerts

Data Warehouse

Snowflake

RAW / SHEETS / SUPABASE · 1.4M+ rows

Transformation

dbt-snowflake 1.11.3

14 models · 49 tests · incremental · documented

Infrastructure

Terraform

AWS + Snowflake · S3 remote state

Cloud

AWS (S3 + SSM)

Parquet storage · SecureString credentials

Containerization

Docker + Docker Hub

mdethereal/supplychain360-airflow:latest

CI/CD

GitHub Actions

flake8 · sqlfluff · dbt test · Docker push · Terraform

Version Control

Git + GitHub

github.com/MDCODE247/supply_chain_plartform

Project Complete

A production-grade, end-to-end data engineering platform built for SupplyChain360 — ingesting, transforming and analysing 1.4M+ rows of supply chain data.

github.com/MDCODE247

- ✅ Data ingestion (S3 · Sheets · Supabase)
- ✅ Parquet + Snowflake loading (1.4M rows)
- ✅ dbt modeling (14 models · 49 tests)
- ✅ Incremental processing
- ✅ Data quality checks (5 checks)
- ✅ Idempotency + failure alerting
- ✅ Terraform IaC (S3 remote state)
- ✅ Docker image on Docker Hub
- ✅ GitHub Actions CI/CD (lint + test)
- ✅ Business insights analytics

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